Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science and Engineering

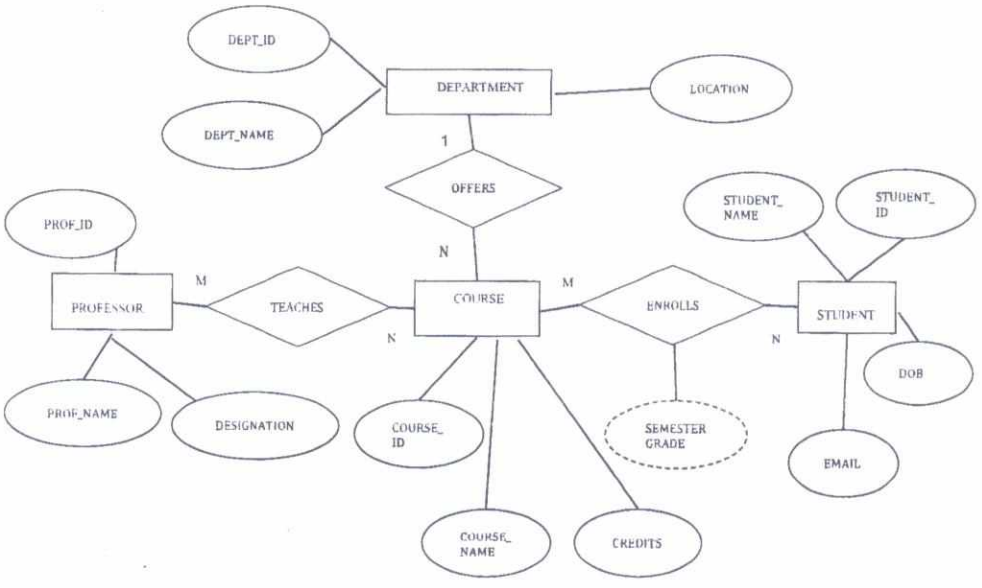
Mid Semester Examination (Summer)

B.E. -Summer Sem	Course Code: UCS310 Course Name: Database Management Systems
Date: June 30 , 2026	Time: 8:00 -10:00
Duration: 2 Hours, M. Marks: 30	Name of Faculty: Dr. Harpreet Singh

Note: Attempt all questions in order. Assume missing data, if any, suitably.

		Marks	CO/ BL
Q1	(a) Consider a hospital that wants to develop a database management system to manage patient records, medical treatments, appointments, and staff details. Different users, such as patients, doctors, nurses, and hospital administrators, require access to specific information while ensuring data security, privacy, and abstraction. Explain how the three-level architecture of DBMS (Internal, Conceptual, and External levels) can be applied to this scenario. (b) Write disadvantages of traditional file based systems	(4+2)	1/2
Q2	(a) Why do we need foreign key constraint? Explain with example (b) Differentiate between alternate key and super key with example (c) Compare and contrast, hierarchical, network and relational database models	(6)	3/2
Q3	Consider the relation: R(A, B, C, D, E, F, G) with the following set of functional dependencies: $A \rightarrow BC$, $B \rightarrow D$, $C \rightarrow E$, $DE \rightarrow F$, $F \rightarrow G$, $G \rightarrow C$, $AB \rightarrow E$, $CE \rightarrow D$ Answer the following: (a) Write algorithm to find canonical cover and closure of attribute sets. (b) Compute the closure of {A} and {CE} (c) Find the canonical cover of the given set of functional dependencies, clearly showing the removal of redundant dependencies and extraneous attributes.	(6)	3/3
Q4	(a) Answer the following : I. What is cardinality of a relation? II. How many tables are needed for a binary relationship with cardinality 1: n ? III. Differentiate between superclass and subclass with example IV. How the schema of a weak entity is specified? V. In a cardinality 1:1, specify the situation where only one table us is needed. VI. Define join dependency.	6	3/2

Q5	(a) How many tables are needed for the following ER diagram? Specify the tables with attributes and primary keys.	4+2	2/3



(b) Describe the tables needed for the following diagram along with constraints.

